

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY**Department: Computer Science and Engineering****Program: B.Sc. in Computer Science and Engineering****Semester Carryover/Clearance/Improvement Examination: Fall 2020****Year: 4th****Semester: 1st****Course Number: CSE4107****Course Name: Artificial Intelligence****Time: 2 (Three) Hours****Full Marks: 50**Use single Answer Script.

Instructions	i)	Answer scripts should be handwritten and should be written in A4 white paper. You must submit the hard copy of the answer script to the Department when the university reopens.
	ii)	You must write the following information at the top page of each part : Department: _____ Program: _____ Course no.: _____ Course Title: _____ Examination: _____ Semester (Session): _____ Student ID: _____ Signature and Date: _____
	iii)	Write down Student ID, Course number and put your signature on top of every single page of the answer script.
	iv)	Write down the page number at the bottom of every page of the answer script.
	v)	Upload the scanned copy of your answer script in PDF format through provided Google Form link within the allocated time. Uploading clear and readable scan copy (uncorrupted) is your responsibility and you must cover all the pages of your answer script. However, for clear and readable scanned copy of the answer script a student should use only one side of a page for answering the questions.
	vi)	You must avoid plagiarism and, maintain academic integrity and ethics . You are not allowed to take any help from another individual. If you do so, it can result in stern disciplinary actions against you from the university authority.
	vii)	Marks allotted are indicated in the right margin .
	viii)	Assume any reasonable data if needed.
	ix)	Symbols and characters have their usual meaning.
	x)	Before uploading the answer script (single PDF file), rename the file as follows. CourseNo_StudentID.pdf Example: CSE4107_160204018.pdf

There are 6 (Six) questions, answer any 4 (Four).

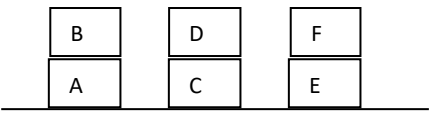
Question 1. [Marks: 12.5]

- a)** Prove or disprove the proposition H using a typical backward chaining algorithm on the knowledgebase that follows.
 $A, B, C, D, A \wedge G \Rightarrow P, A \wedge P \Rightarrow E, C \wedge Q \Rightarrow M, P \wedge R \Rightarrow H, B \wedge E \Rightarrow G, B \wedge D \Rightarrow Q, D \wedge F \Rightarrow G, D \wedge P \Rightarrow F, D \wedge M \Rightarrow P, A \wedge B \Rightarrow R.$ **[4]**
- b)** Analyze the role of proof by contradiction in finding answer to queries to knowledgebases represented in First Order Logic (FOL) through resolution. **[4.5]**
- c)** Present a comparative study between top-down and bottom-up approaches in syntactic analysis of FOL sentences. **[4]**

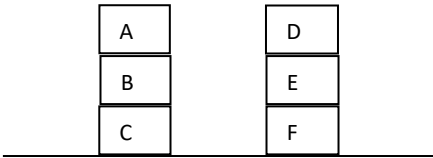
Question 2. [Marks: 12.5]

- a)** Given below through the tables is a weighted undirected graph along with the heuristic function values of the nodes. **[4]**
- | | | | | | | | | | | | | |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|
| Node | A | A | B | B | B | C | C | D | D | E | E | F |
| Neighbor | B | C | C | D | E | E | F | E | G | F | G | G |
| Weight | 46 | 54 | 47 | 36 | 25 | 35 | 32 | 68 | 62 | 57 | 58 | 45 |
-
- | | | | | | | | |
|---------|----|----|----|----|----|----|---|
| Node | A | B | C | D | E | F | G |
| h(Node) | 75 | 51 | 53 | 56 | 54 | 38 | 0 |
- Demonstrate the process of finding a solution to the problem instance taking A as the initial node and using Greedy Best First search.
- b)** Present an analysis of positive and negative sides of local search algorithms. **[4.5]**
- c)** Compare the ideas that underly Best-First graph search and Stochastic Beam search algorithms. **[4]**

Question 3. [Marks: 12.5]

- a)** Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block' and give priority to gain in number of steps over gain in number of actions. **[4]**
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Initial State



Goal State
- b)** Analyze the characteristics of the MINIMAX search strategy applied in game playing. **[4.5]**
- c)** Compare the features of search spaces used for planners and for game playing algorithms. **[4]**

Question 4. [Marks: 12.5]																																	
a)	Full joint-probability distribution of two random variables having three different values each is shown in the table below. <table><tr><td colspan="2" rowspan="2"></td><td colspan="3">X₂</td></tr><tr><td>v₂₁</td><td>v₂₂</td><td>v₂₃</td></tr><tr><td rowspan="3">X₁</td><td>v₁₁</td><td>0.06</td><td>0.07</td><td>0.13</td></tr><tr><td>v₁₂</td><td>0.21</td><td>0.14</td><td>0.12</td></tr><tr><td>v₁₃</td><td>0.08</td><td>0.15</td><td>0.04</td></tr></table> Compute the probabilities of the compound propositions, v₁₃ v₂₃ and v₂₂ v₁₁∨v₁₂ using the distribution.						X ₂			v ₂₁	v ₂₂	v ₂₃	X ₁	v ₁₁	0.06	0.07	0.13	v ₁₂	0.21	0.14	0.12	v ₁₃	0.08	0.15	0.04	[4]							
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b)	Present an analysis of the purpose of Bayesian Networks.				[4.5]																												
c)	Compare the role of chance nodes and decision nodes in influencing the outcome of Decision Networks.				[4]																												
Question 5. [Marks: 12.5]																																	
a)	Consider the ‘Escape the monster’ environment given below. <table><tr><td>OK</td><td>OK</td><td></td></tr><tr><td>1,2</td><td>2,2 S</td><td></td></tr><tr><td>OK</td><td>OK</td><td>OK</td></tr><tr><td>1,1</td><td>2,1</td><td>3,1 S</td></tr></table> Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 64 x 64 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it. Find P(M_{4,1}) taking 0.1 as the independent probability of a monster at any of the unknown cells.				OK	OK		1,2	2,2 S		OK	OK	OK	1,1	2,1	3,1 S	[4]																
OK	OK																																
1,2	2,2 S																																
OK	OK	OK																															
1,1	2,1	3,1 S																															
b)	How do you explain the process of learning that occurs in neural networks functioning based on the backpropagation algorithm?				[4.5]																												
c)	Present a brief comparison between the supervised and unsupervised machine learning strategies.				[4]																												
Question 6. [Marks: 12.5]																																	
a)	Given below is a table that describes some 3-attribute training samples with their class tags. <table><tr><th>A1</th><th>A2</th><th>A3</th><th>Class</th></tr><tr><td>1</td><td>H</td><td>a</td><td>N</td></tr><tr><td>1</td><td>L</td><td>a</td><td>P</td></tr><tr><td>1</td><td>H</td><td>b</td><td>N</td></tr><tr><td>2</td><td>H</td><td>a</td><td>P</td></tr><tr><td>2</td><td>L</td><td>a</td><td>N</td></tr><tr><td>2</td><td>H</td><td>b</td><td>N</td></tr></table> Find the root of the Decision Tree satisfying the samples.				A1	A2	A3	Class	1	H	a	N	1	L	a	P	1	H	b	N	2	H	a	P	2	L	a	N	2	H	b	N	[4]
A1	A2	A3	Class																														
1	H	a	N																														
1	L	a	P																														
1	H	b	N																														
2	H	a	P																														
2	L	a	N																														
2	H	b	N																														
b)	Present an analysis of the idea behind k-Nearest Neighbor classifiers.				[4.5]																												
c)	How do you evaluate the importance of clustering algorithms?				[4]																												